

Drain based Irrigation Scheduling with Offline Reinforcement Learning in Protected Strawberry Cultivation

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Drainage feedback is a practical control signal for substrate-grown strawberries, yet deriving irrigation timing from a drainage target is a sequential decision problem. We present a simulation-anchored offline reinforcement learning approach for protected cultivation in Germany. A FAO56-based evapotranspiration drainage model was calibrated on 107 days of 1 minute tunnel data (481 irrigations) and combined with ten regional weather stations to generate multi-year synthetic episodes. Offline RL agents (BCQ, IQL, PLAS) learned a continuous policy predicting time-to-next irrigation and were evaluated for irrigation-interval versus full-day episode segmentation against a 20% drainage target. Interval episodes produced the best simulated timing (PLAS 80.3% within ± 20 min), while transfer to measured episodes exposed a strong model-to-real gap (37-46% failures).

Keywords

protected horticulture, strawberries, drainage, evapotranspiration, offline reinforcement learning, irrigation scheduling

Protected strawberry production has become a central pillar of Germany's fruit sector. While strawberries are the country's second most important fruit after apples, a growing share of area and with over 15% now comes from protected systems, which provide weather risk mitigation and higher productivity. In 2024, protected cultivation accounted for a substantial fraction of national harvests and has expanded markedly compared to a decade ago (Bundesinformationszentrum Landwirtschaft 2025). At the same time, climate change is amplifying the challenges of irrigation. Rising temperatures, prolonged dry spells and altered rainfall patterns are increasing the overall demand for water and introducing stronger and less predictable fluctuations in daily water requirements. For sensitive crops such as strawberries, this increases the need for responsive irrigation to safeguard both yield and fruit quality (ALFFA 2024).

As a result in substrate-based strawberry production, drainage (the volume and composition of the leachate after irrigation) has emerged as a high-value, high-resolution control signal for adaptive irrigation. Drainage aligns management directly with agronomic goals because it reflects actual plant water uptake, substrate saturation, and nutrient-circulation efficiency (Kubota 2019). Also in drainage-based targets have shown practical leverage: as drainage rate targets increase, applied water rises correspondingly. A 20% drainage rate has been associated with improved vegetative growth and the highest proportion of large fruits, emerging as an appropriate operational target (Su-Hyun Choi 2022). However, converting rich multi-sensor streams, into optimal control decisions is an inherently sequential, uncertainty-laden problem: actions (irrigation events) influence future states across multiple timescales, and decision quality depends on learning from variable and sometimes sparse feedback (Hutchinson 2025).

Reinforcement Learning (RL) offers a principled framework to learn such sequential irrigation policies directly from data and to adapt them to operational constraints, including water scarcity (Yuji Saikai 2023, Xianzhong Ding 2024). Crucially for commercial production, Offline RL can learn effective policies purely from historical operation data, avoiding risky online exploration while still capturing how

climate conditions, drainage responses, and irrigation actions interact over time. This enables data-driven policies that balance plant health, stabilize substrate moisture, and reduce water and fertilizer inputs (T.D. Kelly 2024, Zhao 2023).

The work reported here addresses these gaps through a pragmatic, simulation-anchored offline RL framework targeted at drainage-based scheduling for protected strawberry cultivation in Germany. Specifically, the study (i) calibrates an evapotranspiration-drainage response model for strawberries using an initial season of measured sensor data and subsequently uses public weather station data to generate extended drainage sequences (ii) constructs and trains offline RL agents on the augmented dataset to learn irrigation timing policies (iii) explores two distinct episode segmentation strategies, daily episodes versus session episodes, to evaluate how episode design affects learning and offline policy evaluation.

Materials and Methods

Field Site

An initial trial in an indoor greenhouse (Figure 1, right) was discarded because environmental and irrigation signals exhibited minimal variability and were therefore unsuitable for model development. A subsequent field data acquisition was conducted during the cropping season 2023 on a commercial strawberry (*Fragaria × ananassa*) farm in the Lake Constance region, Germany (47°47' North, 9°30' East, altitude 596m above mean sea level). The mean annual rainfall of the last 10 years is 899.1 mm and average annual mean, maximum and minimum temperatures are 9.8°C, 34.8°C and -11.8°C, respectively. The monitoring happened from 16th May till 14th September including minor interruptions of no more than one week due to work on the plants or damage to the measuring station. The measurement station consists of a 2 m measurement section (Figure 1, left). The cell consisted of two 1 m planting sleeves (substrate mix: 60% coconut fibre, 25% peat and 15% perlite), each containing six plants (Frigo A+, everbearing cultivar Murano) and irrigated by four individual drippers per sleeve. In terms of the cultivation of the plants, they were managed in an identical manner to the surrounding crop by the farm staff. This management included two runner-pruning events with the objective of stimulating renewed vegetative growth. The farm's irrigation management system uses for every irrigation cycle a predetermined duration of 4 minutes and starts at 09:00 AM. Subsequent irrigations were automatically triggered by a timer/radiation-based irrigation control system. These irrigations were subject to continuous supervision and adjustment by staff, who conducted manual drain measurements to ensure optimal hydration levels.



Figure 1: Lake Constance measuring station (left), greenhouse test station (right)

Instrumentation, and Data Collection

A self-developed measuring station (Figure 1, left) recorded air temperature, relative humidity, the applied irrigation water volume (drip), and the drainage volume (drain) directly on the Measurement station. Drip and drain volumes were quantified with tipping-bucket (impulse) rain gauges calibrated to 2 ml per tip. For volume monitoring, one representative dripper was logged for drip, and the combined drain from the entire 2 m section (eight drippers) was collected and measured. Besides the station inside the tunnel there was an outside Sensor (SenseCAP S2102) which recorded the incoming radiation adjacent to the general area.

All variables were logged at a one-minute resolution. The resulting dataset comprises 107 fully monitored days, consisting of 481 irrigation cycles and serves as the foundation for developing the subsequent drainage model.

Estimation of drainage based on crop evapotranspiration:

Drainage was calculated under the assumption that any water applied through irrigation which is not consumed by evapotranspiration contributes directly to drainage. Accordingly, drainage *drain* was derived as the difference between irrigation input *drip* and crop evapotranspiration ET_c :

$$drain = drip - ET_c \quad (Eq. 1)$$

Here, ET_c represents crop evapotranspiration, defined as the combined water loss from soil evaporation and plant transpiration. ET_c was calculated following the widely recognized FAO56 methodology for determining crop water requirements (Richard G. Allen 1998):

$$ET_c = ET_{0\ tunnel} \cdot K_c \quad (Eq. 2)$$

where $ET_{0\ tunnel}$ is the reference evapotranspiration inside the plastic tunnel and K_c is the crop coefficient. Because solar radiation and wind speed are significantly reduced inside plastic-covered tunnels, indoor ET_0 values are known to be lower than standard outdoor reference ET_0 (Pedro Gavilán 2015). To account for this, $ET_{0\ tunnel}$ inside the tunnel was estimated using the method proposed by (M. D. Fernández 2010), which provides empirical equations based solely on the day of the year (DOY) and the internal solar radiation $R_{s\ tunnel}$:

$$DOY < 220: ET_{0\ tunnel} = (0.288 + 0.0019 \cdot DOY) \cdot R_{s\ tunnel} \quad (Eq. 3)$$

$$DOY > 220: ET_{0\ tunnel} = (1.339 - 0.00288 \cdot DOY) \cdot R_{s\ tunnel} \quad (Eq. 4)$$

Despite being derived for Mediterranean climates, the deviation from local conditions is expected to be compensated by the later calibration of the model using sensor data. The required internal solar radiation $R_{s\ tunnel}$ was estimated from measured outdoor solar radiation R_s and the transmissivity τ of the plastic cover:

$$R_{s\ tunnel} = R_s \cdot \tau \quad (Eq. 5)$$

The transmissivity depends on the tunnel covering material, its age, and its cleanliness therefore a value of $\tau = 0.75$, reported by Lozano et al. (David Lozano 2016), was used as it reflects typical conditions of the polytunnels on the farm.

The crop coefficient K_c from (Eq. 2) adjusts reference evapotranspiration to the specific crop conditions. Published K_c values for strawberries in southern Germany are in the range [0.6,0.9] (Martin

Müller 2020), but these refer exclusively to open-field cultivation. In tunnel-grown remontant strawberries, additional factors significantly alter water use, including the two runner-pruning events mentioned earlier, which substantially reduce transpiration. Furthermore, ET_c calculations are typically referenced to 1 m² of ground area, whereas our measurement cell consists of a strawberry row in substrate bags with an unrepresentative area, requiring an additional area-scaling factor. All such uncertainties were consolidated into the K_c parameter and expressed as a time-dependent factor varying on a weekly basis and calibrated with the collected datasets.

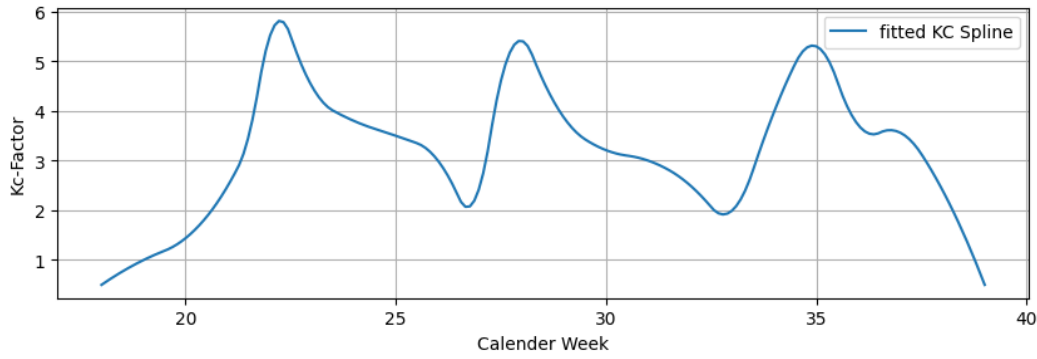


Figure 2: Kc Factor by week

Synthetic drainage generation

To generate training datasets, meteorological data from the past five years were used, originating from ten public weather stations operated by the (Landwirtschaftliches Technologiezentrum Augustenberg 2026). The selected stations were located near the study site and provided measurements of air temperature, relative humidity, and solar radiation at a resolution of 10 minutes. Only data from calendar weeks 18 to 39 were included, as irrigation decisions must be made exclusively within this period. Synthetic drainage was computed using solar radiation data from the weather stations in combination with the assumptions defined in (Eq. 1). The irrigation volume per event was fixed due to a constant irrigation duration of 4 minutes and validated through sensor data analysis, resulting in an average applied volume of 5 l per irrigation event. To introduce realistic variability into the training data, a target drainage value was randomly sampled within an allowable deviation range. The ideal drainage/drip ratio was set to be 20% according to (Su-Hyun Choi 2022), corresponding to 1 l. A maximum deviation of 150% relative to this ideal value was permitted in the datasets which results in a theoretical range of [-0.5, 2.5] l. The selected random value from this range defines the target drainage $\text{drain}_{\text{target}}$ for an irrigation interval. For each interval, the model iterates forward from the last irrigation event, accumulating evapotranspiration $ET_{c,\text{cumulativ}}$ starting at zero. The interval progresses until (according to (Eq. 1)):

$$ET_{c,\text{cumulativ}} \geq \text{drip} - \text{drain}_{\text{target}} \quad (\text{Eq. 6})$$

is reached and the interval is terminated. This time point represents the moment at which a subsequent irrigation event would produce exactly the selected $\text{drain}_{\text{target}}$. This procedure ensures internal consistency: every generated drainage value is physically linked to the evapotranspiration dynamics of the respective episode. Because $\text{drain}_{\text{target}}$ is a scalar but model inputs require a time series, empirical drainage profiles derived from measurements were imposed such that 70% of the total drainage occurs 10 minutes after irrigation (first sample), 90% after 20 minutes (second sample), and 100% at all subsequent timesteps.

Offline Reinforcement Learning

Offline RL, also known as batch RL, is a learning paradigm in which an agent is trained using a fixed, previously recorded dataset. During the learning process, the agent is not permitted to interact with the real or simulated environment. This differs fundamentally from classical reinforcement learning, in which an agent actively explores to gradually learn a model of the environment and an optimal strategy. It aims to derive policies that improve over the behavior policy of the original system while avoiding unsafe extrapolation beyond the data distribution. A key technical difficulty is that Offline RL must estimate policies for states and actions that may not be adequately represented in the dataset. Formally, this dataset consists of transition tuples (Sergey Levine 2020):

$$\mathcal{D} = \{(s_i, a_i, s'_i, r_i)\} \quad (\text{Eq. 7})$$

where s_i is the current state, a_i the applied action, r_i the reward, and s'_i the next state. This setting avoids risky online exploration and leverages historical operation data for policy learning.

The state s is designed to capture the relevant environmental and system conditions at a given decision time, while remaining low-dimensional enough to allow efficient learning. We define the state vector as:

$$s = \{d, t_{abs}, t_{irri}, T, H, R, D\} \quad (\text{Eq. 8})$$

with d = day-of-year, t_{abs} = absolute time of day, t_{irri} = time since the most recent irrigation and T, H, R, D = cumulative temperature, relative humidity, solar radiation and drainage. This State reflects thereby both the seasonal and daily context alongside cumulative atmospheric demand and the system's hydrologic response.

The action a denotes the set of admissible decisions available to the agent in each state. The action space is continuous and parametrized as the predicted time until the next irrigation event. For each transition, the model outputs a real-valued scalar indicating how many minutes remain until the next irrigation should occur.

$$a \in \mathbb{R} \quad (\text{Eq. 9})$$

The reward r is defined as the function $r(\text{drain}_{target})$, and assigns a scalar reward for each irrigation interval based on a given drain_{target} . As illustrated in Figure 3 a piecewise-linear, clipped reward is used. It is positive inside the tolerance band $[0.5, 1.5]$ l based on $\pm 10\%$ of the absolute irrigation amount and negative outside, peaking at $\text{drain}_{target} = 1$ l corresponding to the 20% drainage.

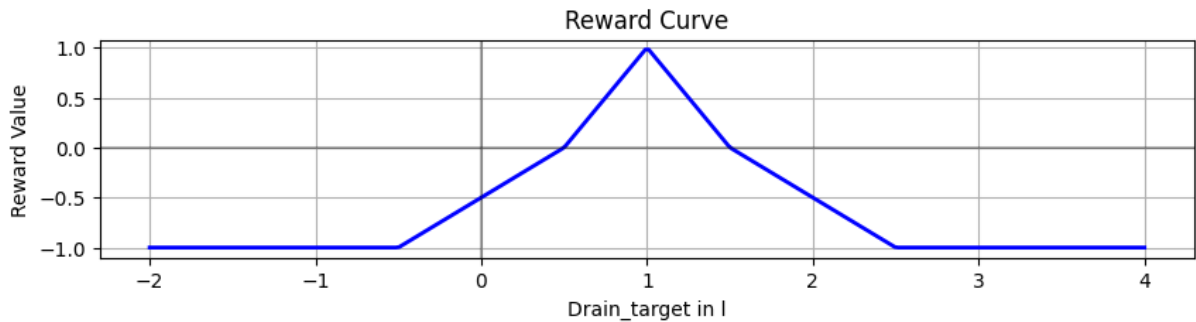


Figure 3: Reward curve $r(\text{drain}_{target})$

Prior to training, both the scalar action a and state s variables were linearly normalized using min-max preprocessing, ensuring that all actions and rewards were scaled to the interval $[-1,1]$. For the training two different episode-construction strategies were evaluated.

- The first approach followed an irrigation interval based episode structure in which all cumulative environmental variables (T, H, R, D) were reset at each irrigation event to capture only the short-term atmospheric demand and system response. The final episode of the dataset was excluded, as it never terminated naturally and was found to negatively impact training stability.
- The second approach instead used a full day based episode structure starting at 09:00 AM (from the first irrigation) till midnight, without resetting cumulative variables. Each episode therefore had a fixed length determined by the predefined start/endpoint and sampling frequency.

For the training only the model-generated dataset is used, while the recorded measurements will be reserved exclusively for evaluation to ensure a clean separation between training and assessment. To conduct the training, the open-source offline deep reinforcement learning library d3rlpy (Imai 2022) was employed. This library provides a broad set of state-of-the-art algorithms and a user-friendly interface. To ensure a rigorous and unbiased algorithm selection, an initial benchmark run using all algorithms available in d3rlpy with default hyperparameters was conducted. The three best-performing algorithms were then selected based on the action-difference metric between dataset actions and agent-generated actions. For both approaches the top performers are policy in the latent action space (PLAS) (Held 2020), batch-constrained Q-learning (BCQ) (Precup 2019), and implicit Q-learning (IQL) (I. K. Levine 2021).

Results

Drainage Model

The evapotranspiration-driven drainage model was evaluated by comparing daily calculated evaporation values against measured daily evaporation inferred from the on-site irrigation drainage balance. The comparison is shown in Figure 4, where each marker represents one day of the monitoring campaign and is complemented by an ideal reference line and a linear regression fit. Overall, the point cloud indicates a clear positive association between calculated and measured values, while also revealing a non-negligible spread around the ideal line, suggesting a remaining model and measurement uncertainty at the daily aggregation level. This observation is further supported by the model metrics reported in Table 1.

Table 1: Determination coefficient (R^2), root mean square error (RMSE), relative error (RE), mean absolute error (MAE) and mean bias error (MBE) from the comparison between calculated versus measured evaporation values.

R^2	RMSE in l day^{-1}	RE in %	MAE in l day^{-1}	MBE in l day^{-1}
0.69	13.21	36.8	9.79	4.15

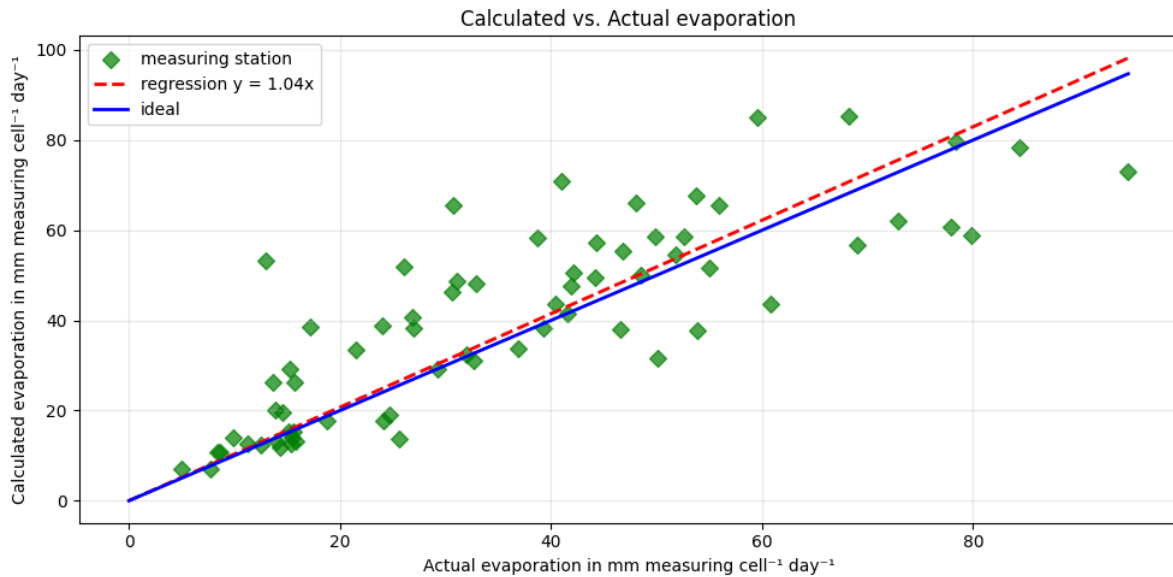


Figure 4: Comparison between calculated versus measured evaporation values. Each point corresponds with a daily evaporation value. Also shown is the 1:1 line and a linear regression line.

Training results

Figure 5 illustrates representative irrigation episodes by plotting the predicted action over the course of the interval for all three learned policies against the ideal schedule derived from the simulated drainage-target termination criterion. Because the environment and dataset are available at a discrete sampling rate of 10 minutes, policy outputs were translated into a discrete irrigation decision using a 10-minute execution threshold. Whenever a policy predicted a remaining time-to-irrigation of smaller than 10 minutes, the subsequent sample was treated as the executed irrigation event. This discretized irrigation timestamp is then used as the predicted irrigation time for further evaluations.

The qualitative behavior in Figure 5 already highlights clear differences between algorithms and episode constructions. For irrigation-interval episodes (Figure 5a), algorithms typically follow the characteristic decline of the ideal irrigation prediction curve. For full-day episodes (Figure 5b), all algorithms exhibit less structured action trajectories. To quantify this action accuracy from the predictions of each algorithm and approach, performance was assessed by the timing offset between the predicted and the ideal irrigation event for every episode. Where negative values indicate earlier irrigation than the reference schedule and positive values indicate late irrigation. These results were then visually represented using a violin plot overlaid on a box plot in Figure 6. In addition, we report the fraction of episodes for which a policy did not trigger any irrigation within a tolerance window of ± 90 min (reported as ‘Errors’ in Table 2), which captures failure cases caused by overly conservative actions or oscillatory predictions that never cross the execution threshold.

Considering the quantitative results on the training dataset (Table 2 top, sim and Figure 6a, left column), the irrigation-interval formulation shows that PLAS achieves the best timing accuracy (± 10 min: 58.59%, ± 20 min: 80.30%, ± 30 min: 89.61%). At the same time, it yields very few failures (1.17%) and the tightest standard distribution (StdDev) of 19.85 min. IQL performs comparably, reaching 51.24% / 73.88% / 84.91% within the same tolerance windows, with a 1.70% failure rate (StdDev: 24.73 min). In contrast, BCQ trails substantially (11.29% / 24.31% / 40.91%) and exhibits a much wider dispersion (StdDev: 43.07 min), aligning with the more erratic trajectories observed in Figure 5a.

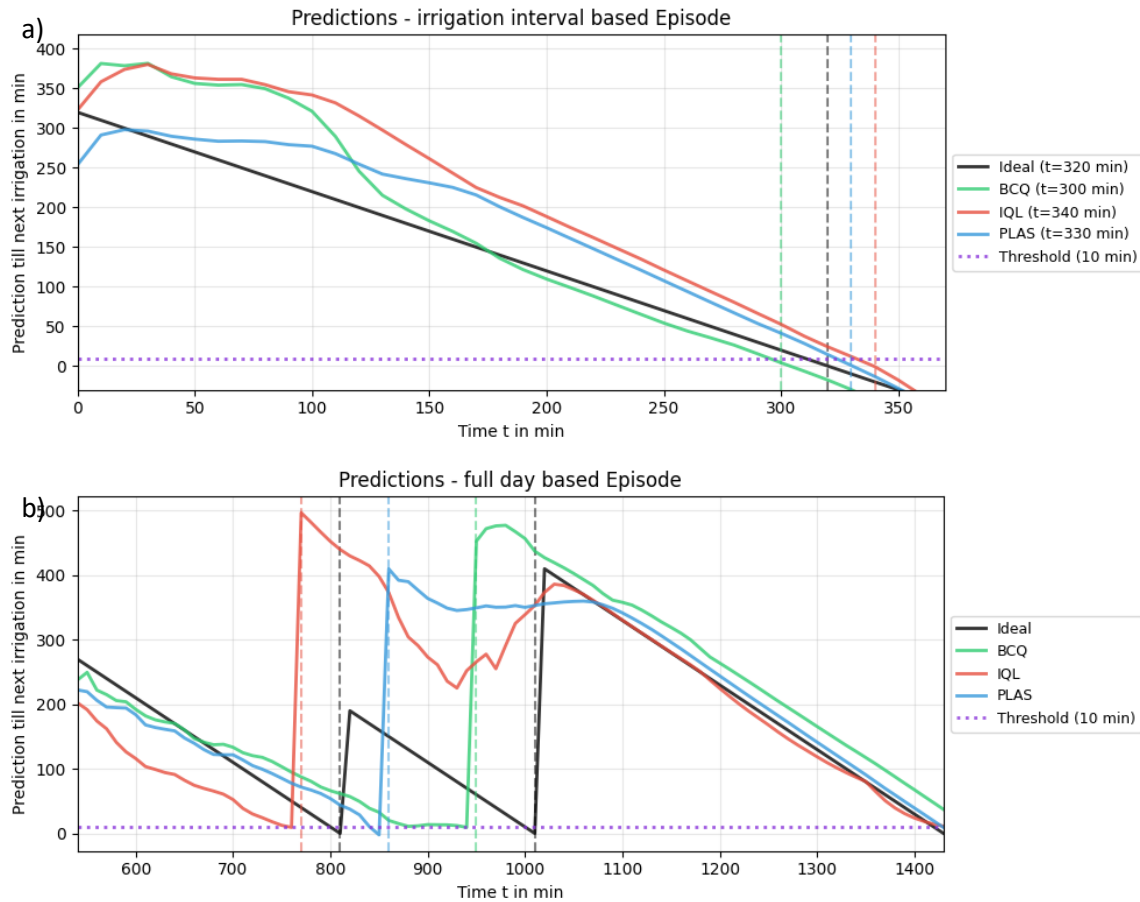
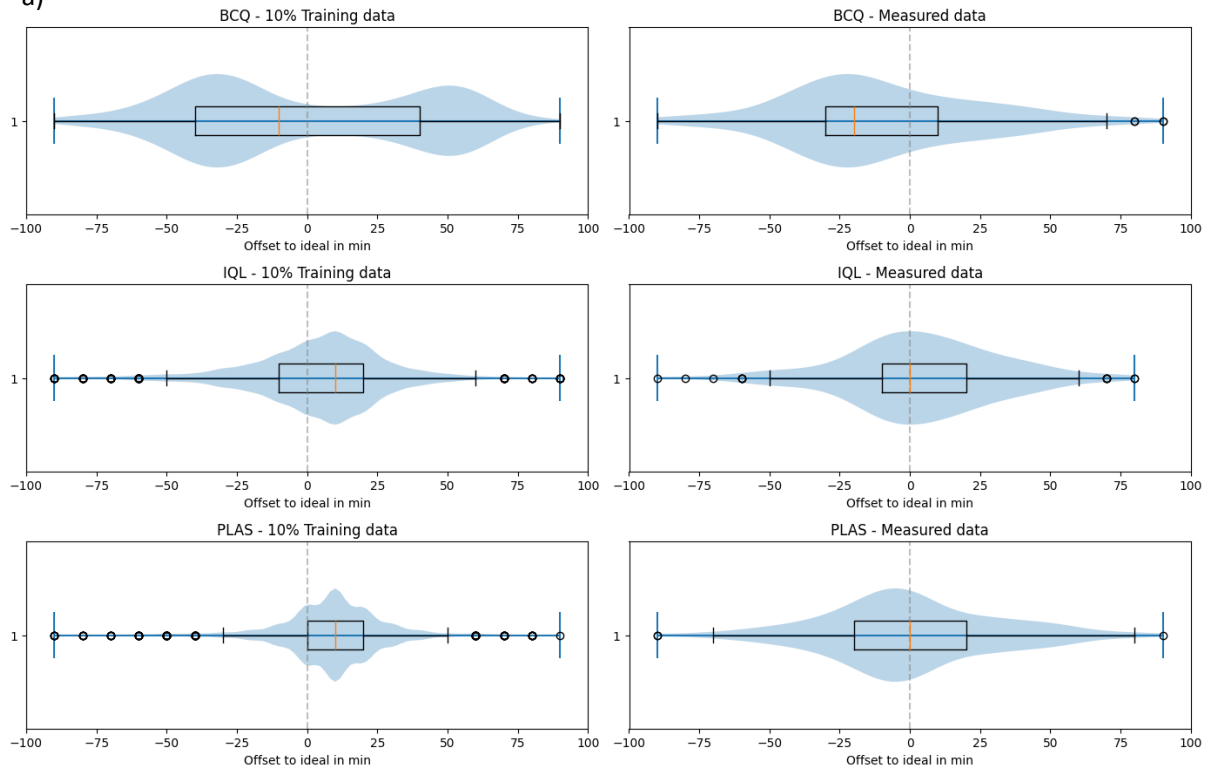


Figure 5: Example episodes showing the predicted action over time for three offline RL policies (BCQ, IQL, PLAS) compared to the ideal irrigation schedule. a) for the irrigation interval based Episode and b) for the full day based Episode. The solid black curve indicates the ideal action and the colored curves correspond to the respective policy predictions. The vertical dashed lines indicate when each policy would irrigate.

When transferred to real measurement episodes (Table 2 top, meas and Figure 6a, right column), performance deteriorated markedly for all policies, revealing a pronounced model-to-real gap. For irrigation-interval episodes, the proportion of failures ($> \pm 90$ min) increased to 37.27-46.14%, and the share of predictions within operationally relevant tolerances dropped to 30.68-37.27% (± 20 min) and 40.68-42.95% (± 30 min). The offset distributions also widen considerably, indicating reduced confidence and higher episode-to-episode variability. BCQ shows a tendency towards earlier irrigation on average (mean -9.96 min), whereas IQL and PLAS remain closer to unbiased timing (mean $+4.22$ min and $+0.17$ min, respectively) but still with sizeable spread (StdDev 28.53-31.11 min).

Considering the full-day based episode approach (Table 2 bottom, sim and Figure 6b, left column), the timing-offset distributions are substantially worse than for the irrigation-interval based approach. The results indicate that BCQ and PLAS frequently fail to produce a valid irrigation decision within the ± 90 min evaluation window, with high failure rates of 41.08% and 47.31%, respectively, and correspondingly low timing accuracy. IQL remains the best-performing method under this episode design, but it still exhibits 12.92% failures and achieves only 57.85% of predictions within ± 20 minutes of the ideal schedule. When moving to measured conditions (Table 2 bottom, meas and Figure 6b, right column), performance degrades even further, effectively resulting in a total breakdown of the full-day approach. All three algorithms show extremely high error rates between 89.91% and 99.12%, meaning that in most days no irrigation time is identified within ± 90 minutes. Consistently, the models virtually never predict irrigation within ± 30 minutes of the ideal reference under real measurements.

a)



b)

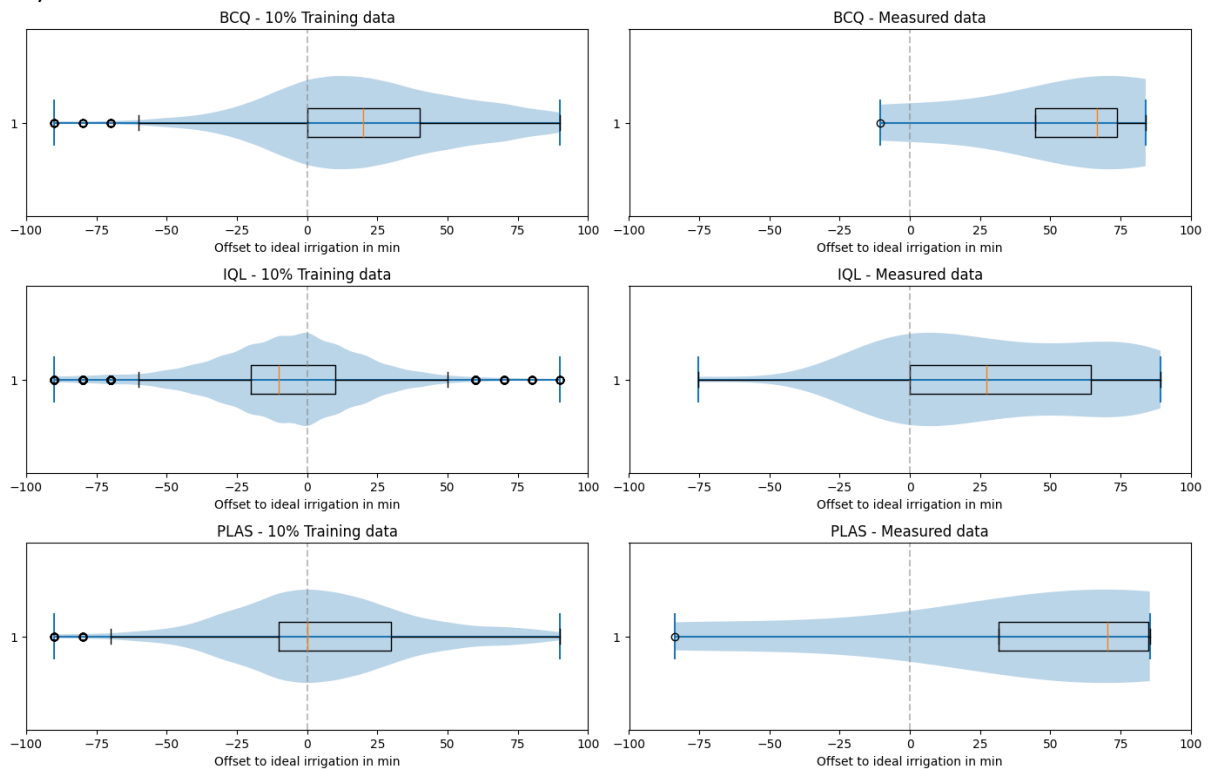


Figure 6: Distributions of the offset to the ideal schedule (minutes) for three methods (BCQ, IQL, PLAS) under test data (left column) and measured (right column) conditions (excluding predictions $>\pm 90$ min). a) for the irrigation interval based Episode and b) for the full day based Episode. Each panel shows a violin density with an overlaid box (median and interquartile range) and individual outliers (circles), the vertical dashed line marks 0 min (no offset).

Table 2: Evaluation results of the trained models on training data (sim) and real measurement data (meas). Errors refers to the proportion of episodes in which the model did not identify an irrigation time in a range of ± 90 min. The columns within $\pm 10/20/30$ min indicate the percentage of episodes in which the time deviation between the predicted and ideal irrigation times is within the respective tolerance range. The statistical indicators mean and standard deviation (StdDev) describe the distribution of the time deviation in minutes

Model	Number of Episodes	Errors ($>\pm 90$ min) in %	Within ± 10 min in %	Within ± 20 min in %	Within ± 30 min in %	Mean Offset in min	StdDev in min
Irrigation intervall based Episodes:							
BCQ - sim	3580	6.06	11.29	24.31	40.91	-1.68	43.07
IQL - sim	3580	1.7	51.24	73.88	84.91	3.93	24.73
PLAS - sim	3580	1.17	58.59	80.3	89.61	8.58	19.85
BCQ - meas	481	37.27	15.45	30.68	41.59	-9.96	34.7
IQL - meas	481	44.55	26.36	37.27	42.95	4.22	28.53
PLAS - meas	481	46.14	25.0	33.18	40.68	0.17	31.11
Full day based Episodes:							
BCQ - sim	1142	41.08	18.97	29.13	37.11	20.79	33.81
IQL - sim	1142	12.92	39.32	57.85	69.29	-7.2	28.93
PLAS - sim	1142	47.31	20.49	30.82	38.53	6.2	33.55
BCQ - meas	107	89.91	1.1	3.29	4.82	30.66	38.53
IQL - meas	107	98.9	0.0	0.0	0.0	37.64	63.85
PLAS - meas	107	99.12	0.0	0.22	0.22	51.69	36.67

Overall, the irrigation-interval episode approach enables strong learning performance with PLAS and IQL achieving high timing accuracy and low error rates. However, evaluation on measured episodes reveals a pronounced model-to-real gap characterized by a systematic earlier-irrigation bias across all algorithms. The full-day episode approach proved considerably less effective. Even on training data the algorithms show high failure rates. On measured full-day episodes, all three methods fail totally. This indicates that learning a stable policy over long, non-resetting horizons is substantially harder in the current state representation and dataset, and that the interval-based segmentation is the more appropriate design for drainage-targeted scheduling with the available data.

Discussion

Interpreting the evapotranspiration - drainage model

The drainage model reaches a determination coefficient of $R^2 = 0.69$ for daily evapotranspiration estimates, indicating that radiation-driven ET captures a substantial portion of day-to-day water demand but leaves meaningful unexplained variance. This is most likely because the model compresses multiple uncertainties into a time-varying weekly crop coefficient (K_C), including plant development, runner-pruning events, and the mismatch between ground-area reference ET_C and the effective area of substrate bags. While this consolidation is pragmatic, it also means that K_C absorbs phenomena that may change abruptly (e.g. runner pruning), which a weekly schedule may only approximate. In addition, the tunnel microclimate differs from nearby weather stations and even from a single point inside the tunnel, spatial gradients in radiation, humidity, and temperature can be large in protected horticulture. Despite these limitations, the model appears sufficiently informative to learn policies in-distribution, as evidenced by the strong results on simulated episodes. Importantly, the observed bias

and spread are consistent with a model that is heading in the right direction but is far from a digital twin.

Offline reinforcement learning performance: algorithms and episode design

On simulated data, the irrigation-interval episode design clearly outperforms the full-day design. Resetting cumulative variables at each irrigation event likely simplifies the learning problem by focusing the state on the local decision context (recent atmospheric demand and recent drainage response) instead of requiring the agent to integrate long-horizon dynamics over an entire day. With the current state representation, the full-day formulation appears partially muted, which may be due to the fact that cumulative variables grow monotonically and can blur the distinction between different intra-day cycles, increasing function-approximation difficulty for value learning.

The discretization of a continuous action into a 10 minute execution threshold is a practical choice given the 10 minute sampling of the synthetic environment. However, it introduces an additional source of evaluation noise: two policies that differ only slightly around the threshold can trigger at different timesteps. Future implementations could instead formulate the action directly as a discrete decision at each timestep (irrigate now vs wait) or increase temporal resolution where sensor and actuation hardware allow.

Understanding the model-to-real gap

Policies that are highly accurate on simulated intervals deteriorate sharply on measured episodes, with a large fraction of intervals receiving no valid irrigation decision within ± 90 minutes. This gap likely arises from a combination of the already mentioned model mismatch, reward mismatch, and distribution shift in the state action coverage. The reward mismatch is present because the ideal schedule in the training dataset is defined by a criterion that exactly achieves the sampled drainage target, but real farm scheduling is timer/radiation controlled and periodically adjusted by staff. Therefore, the measured actions may reflect operational heuristics, equipment constraints, and risk aversion rather than strict drainage-optimality. When a policy is trained to imitate synthetic optimal timing but evaluated against real episodes or vice versa, apparent performance can degrade even if the policy is agronomically reasonable. The coverage shift is like present because offline RL depends critically on whether the evaluation states lie within the support of the training dataset. Even with min-max normalization, measured episodes may present combinations that are rare or absent in the synthetic data, especially around atypical weather patterns, irrigation disruptions, or post-pruning recovery. Under such shift, Q-function extrapolation becomes unreliable and conservative algorithms may default to wait, producing the observed failure modes.

The failure of the full-day approach under measured conditions further suggests that long-term policies are particularly sensitive to mismatch. Errors accumulate across the day because the state is not reset, and any early misprediction pushes subsequent states further out of distribution. This highlights the practical value of designing episodes that align with meaningful, recurrent control cycles - in this case, irrigation intervals.

Implications and recommendations for future work

For deployment-oriented irrigation control, several concrete improvements follow from the results.

Expand real-world sensing: adding substrate moisture/nutrient content, solution nutrient content and ideally weight-based lysimeter signals would reduce partial observability and allow the agent to condition decisions on the substrate's actual water status rather than proxies alone.

Improve model fidelity and diversity: calibrate transmissivity and microclimate relationships per tunnel and season. Apply domain randomization (e.g., varying K_C trajectories, irrigation volumes, drainage delay parameters) so that the synthetic dataset spans plausible real dynamics.

Use safer offline RL and validation: incorporate uncertainty estimation or conservative penalties to reduce extrapolation, complement action-difference metrics with off-policy evaluation estimators, and test policies in shadow mode before allowing actuation.

Overall, the present study demonstrates that offline RL can learn precise interval-level irrigation timing under a well-defined synthetic objective, but it also makes clear that data quantity, state observability, and model calibration dominate success when transitioning to real protected horticulture systems.

Conclusions

In summary, this paper shows that drainage-targeted irrigation scheduling is a promising application for offline reinforcement learning, but robust real-world performance requires closing the model to reality gap.

- A radiation-driven evapotranspiration model calibrated on 107 monitored days provides a workable drainage simulator for synthetic data generation, achieving $R^2 = 0.69$ at daily scale.
- Offline RL can learn irrigation timing policies that closely match a drainage-target ideal schedule in simulation when episodes are segmented by irrigation intervals, PLAS and IQL are markedly more stable than BCQ under this formulation.
- Full-day episode segmentation performs substantially worse and becomes brittle under measured conditions, indicating that long, non-resetting horizons are challenging with the current low-dimensional state representation.
- A pronounced model-to-real gap persists. Policies trained on synthetic data degrade strongly on measured episodes, highlighting model mismatch and distribution shift as the central barriers to field deployment.
- Future work should prioritize richer in-tunnel sensing, model randomization, and conservative validation workflows (e.g., shadow deployment) to enable safe, drainage-based RL irrigation control.

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